



Semantic Web

Lab 3 – Java

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Lab CONTENT

Java Inheritance **01**

Java Exception Handling **02**

INHERITANCE

- ❑ Save time during program development
- ❑ The existing class is called the superclass, and the new class is the subclass.
- ❑ A subclass is more specific than its superclass
- ❑ In Java, the class hierarchy begins with class Object.

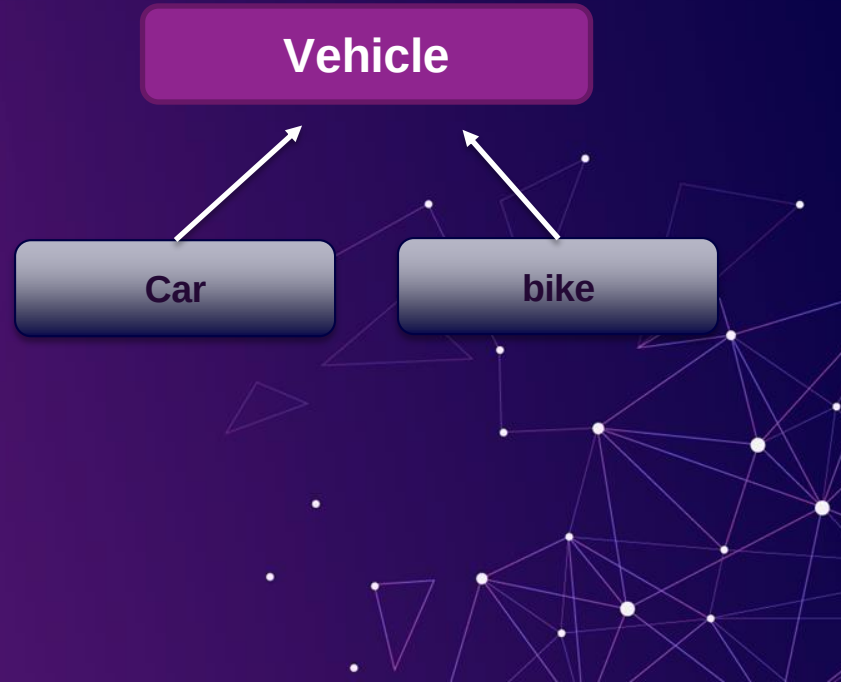


INHERITANCE...

- Class **Car** be a subclass of class **Vehicle**

Every **Vehicle** is a **Car** ?

- One superclass can have many subclasses



Inheritance Example

```
class Car extends Vehicle {  
  
    public static void main(String[] args) {  
        Car myFastCar = new Car();  
        myFastCar.VehicleModel("2015");  
    }  
}  
  
class Vehicle {  
    protected String brand = "Ford";  
  
    public void VehicleModel(String model) {  
        System.out.println("brand : "+brand);  
        System.out.println("model : "+model);  
    }  
}
```

```
brand : Ford  
model : 2015
```

CLASS MEMBERS (**ACCESS MODIFIERS**)

❑ **A class's public members ??**

are accessible wherever the program has a reference to an object of that class or one of its subclasses.

❑ **A class's private members ??**

are accessible only within the class itself.

❑ **A class's protected members ??**

- A superclass's protected members can be accessed by members of that superclass.
- by members of its subclasses .
- by members of other classes in the same package.

RELATIONSHIP BETWEEN SUPER CLASSES & SUBCLASSES

Example : We now use an inheritance hierarchy containing types of employees in a company's payroll application to discuss the relationship between a superclass and its subclass.



Java Program..

```
public class CommissionEmployee {  
    private String firstName;  
    private String lastName;  
    private String SSN;  
    private double grossSalary;  
    private double commissionRate;  
  
    public CommissionEmployee(String first , String last , String ssn , double gs , double cr){  
        this.firstName      = first;  
        this.lastName        = last;  
        this.SSN             = ssn;  
        this.grossSalary     = gs;  
        this.commissionRate  = cr;  
    }  
}
```


Java Program..

```
public void setFirstName(String first){  
    firstName = first;  
}  
public String getFirstName(){  
    return firstName;  
}
```

```
public void setLastName(String last){  
    lastName = last;  
}  
public String getLastName(){  
    return lastName;  
}
```

```
public void setSSN(String ssn){  
    SSN = ssn;  
}
```

```
public String getSSN(){  
    return SSN;
```

```
}
```

Java Program..

```
public void setGrossSalary(double gs){
    if(gs > 0.0){
        grossSalary = gs;
    }else{
        System.out.println("Gross Salary must > 0");
    }
}

public double getGrossSalary(){
    return grossSalary;
}

public void setCommissionRate(double rate){
    if(rate >0.0 && rate <1.0){
        commissionRate = rate;
    }else{
        System.out.println("rate must > 0 && less than 1");
    }
}

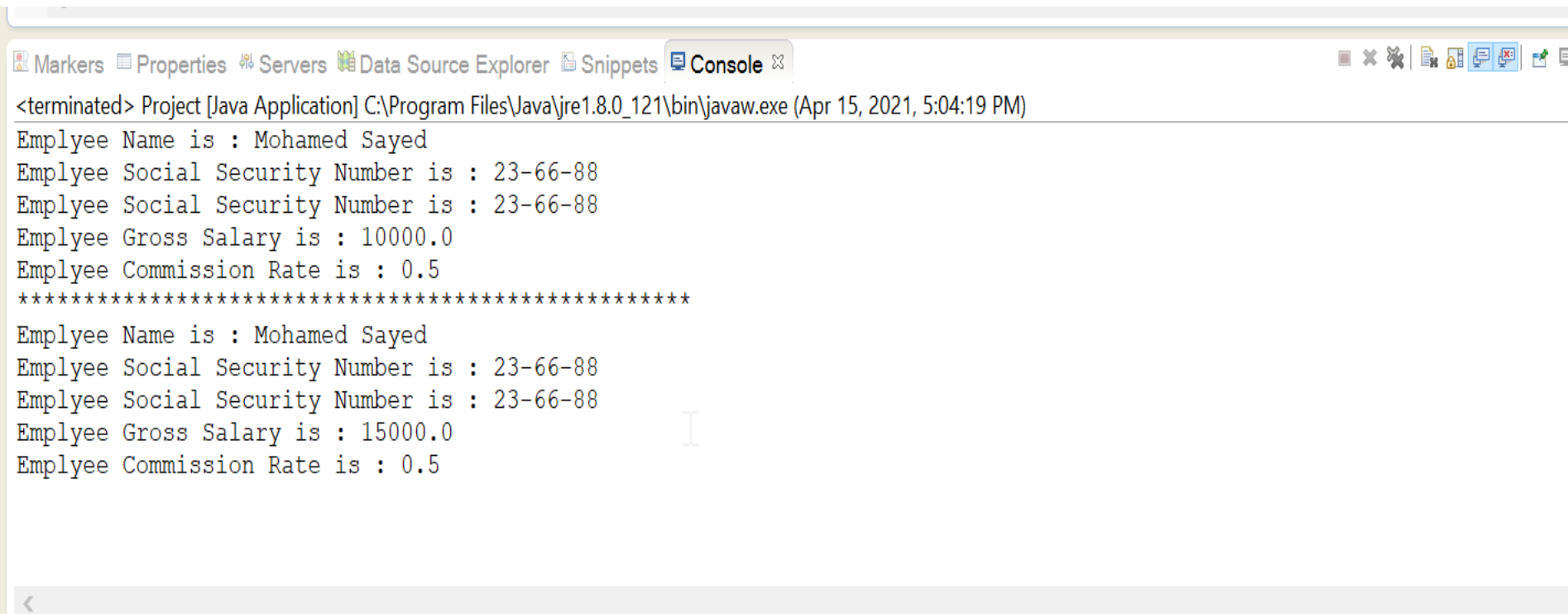
public double getCommissionRate(){
    return commissionRate;
}
```

Java Program..

```
public void PrintEmployeeInfo() {  
    System.out.println("Employee Name is : " + firstName + " " + lastName);  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Gross Salary is : " + grossSalary );  
    System.out.println("Employee Commission Rate is : " + commissionRate );  
}
```

```
1  
  
public class Project {  
  
    public static void main(String[] args) {  
  
        CommissionEmployee emplyess = new CommissionEmployee("Mohamed" , "Sayed" , "23-66-88" , 10000 , 0.5);  
        emplyess.PrintEmployeeInfo();  
  
        // Make Update  
        emplyess.setGrossSalary(15000);  
        emplyess.PrintEmployeeInfo();  
  
    }  
}
```

Output



The screenshot shows an IDE's console window with a toolbar at the top containing icons for Markers, Properties, Servers, Data Source Explorer, Snippets, and Console. The console title bar reads "Console". The output text is as follows:

```
<terminated> Project [Java Application] C:\Program Files\Java\jre1.8.0_121\bin\javaw.exe (Apr 15, 2021, 5:04:19 PM)
Employee Name is : Mohamed Sayed
Employee Social Security Number is : 23-66-88
Employee Social Security Number is : 23-66-88
Employee Gross Salary is : 10000.0
Employee Commission Rate is : 0.5
*****
Employee Name is : Mohamed Sayed
Employee Social Security Number is : 23-66-88
Employee Social Security Number is : 23-66-88
Employee Gross Salary is : 15000.0
Employee Commission Rate is : 0.5
```

A vertical cursor is positioned between the two sets of output. At the bottom of the console, there is a light gray bar with a left-pointing arrow icon.

NOTICE !!

- ❑ Much of class BasePlusCommissionEmployee's code is similar, or identical, to that of class CommissionEmployee.
- ❑ We literally copied code from class CommissionEmployee and pasted it into class BasePlusCommissionEmployee
- ❑ This “copy-and-paste” approach is often error prone and time consuming. Worse yet, it spreads copies of the same code throughout a system, creating a code-maintenance nightmare

Solution ???

Java Program..

```
public class BasePlusCommissionEmployee extends CommissionEmployee{  
  
    private double baseSalary;  
  
    public BasePlusCommissionEmployee(String first , String last , String ssn ,  
        double gs , double cr , double bs){  
  
        super(first, last , ssn , gs , cr ); // Constructor of CommissionEmployee  
  
        this.baseSalary = bs;  
    }  
}
```

Java Program..

```
public void setBaseSalary(double bs){  
    if(bs > 0.0){  
        baseSalary = bs;  
    }else{  
        System.out.println("Base Salary must > 0");  
    }  
}  
  
public double getBaseSalary(){  
    return baseSalary;  
}
```

Java Program..

```
public void PrintEmployeeInfo()  
{  
    System.out.println("Employee Name : " + firstName + " " + lastName);  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Gross Salary : " + grossSalary );  
    System.out.println("Employee Commission Rate : " + commissionRate );  
}
```

```
public class CommissionEmployee {  
    protected String firstName;  
    protected String lastName;  
    protected String SSN;  
    protected double grossSalary;  
    protected double commissionRate;  
}
```


Java Program..

```
public void PrintEmployeeInfo() {  
    System.out.println("Employee Name is : " + firstName + " " + lastName);  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Social Security Number is : " + SSN );  
    System.out.println("Employee Gross Salary is : " + grossSalary );  
    System.out.println("Employee Base Salary is : " + baseSalary );  
    System.out.println("Employee Commission Rate is : " + commissionRate );  
}
```

```
public static void main(String[] args) {
```

```
/* CommissionEmployee employess = new CommissionEmployee("Mohamed" , "Sayed" , "23-66-88" , 10000 , 0.5);  
employess.PrintEmployeeInfo();
```

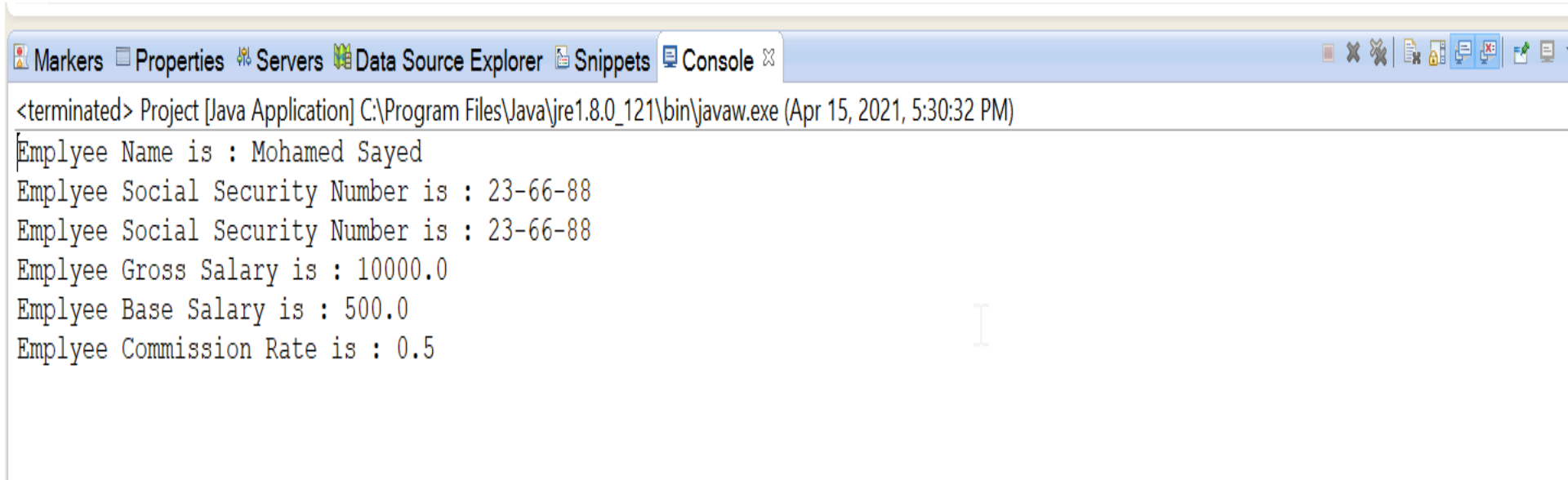
```
System.out.println("*****");  
// Make Update  
employess.setGrossSalary(15000);  
employess.PrintEmployeeInfo();*/
```

```
BasePlusCommissionEmployee plusEmployee = new BasePlusCommissionEmployee("Mohamed" , "Sayed" , "23-66-88"  
    , 10000 , 0.5 , 500);
```

```
plusEmployee.PrintEmployeeInfo();
```

```
}
```

Output



The screenshot shows an IDE window with a console tab active. The console displays the output of a Java application. The output consists of six lines of text, each representing a different attribute of an employee. The first line indicates the program has terminated. The subsequent lines show the employee's name, social security number (displayed twice), gross salary, base salary, and commission rate. The IDE interface includes a toolbar with icons for Markers, Properties, Servers, Data Source Explorer, Snippets, and Console. The console title bar indicates the project is a Java Application and provides the path to the Java runtime executable.

```
<terminated> Project [Java Application] C:\Program Files\Java\jre1.8.0_121\bin\javaw.exe (Apr 15, 2021, 5:30:32 PM)
Employee Name is : Mohamed Sayed
Employee Social Security Number is : 23-66-88
Employee Social Security Number is : 23-66-88
Employee Gross Salary is : 10000.0
Employee Base Salary is : 500.0
Employee Commission Rate is : 0.5
```

Exception Handling

```
public class JavaExceptionExample{  
    public static void main(String args[]){  
        try{  
            //code that may raise exception  
            int data=100/0;  
        }catch(ArithmeticException e){  
            System.out.println(e);}  
            //rest code of the program  
            System.out.println("rest of the code...");  
        }  
    }
```



THANKS

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